WEEK 2 PRACTICUM

COURSE: CYB

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TransitShield: An Interactive ODTN Cyberattack Crisis Simulator

I have selected ODTN Scenario: Cyberattack Cripples Transit – Commuters Stranded

Project Title: TransitShield: An Interactive ODTN Cyberattack Crisis Simulator

Intended Audience: Cybersecurity students, transit employees, IT professionals, and people interested in incident response and network security.

1. Introduction

For this assignment, I developed an interactive cybersecurity crisis simulator called TransitShield, based on the ODTN scenario "Cyberattack Cripples Transit – Commuters Stranded." The project demonstrates how a cyberattack can disrupt public transportation systems and affect commuters, employees, and the daily operations of a transit agency.

The purpose of the website is to help users understand how an organization should respond to a cybersecurity incident. Users take the role of an incident response team member and make decisions about containing the cyberattack, protecting passengers, investigating the incident, communicating with the public, and restoring transit services safely.

I chose this project because it combines cybersecurity knowledge with practical decision-making. Instead of only reading about a cyberattack, users participate in a simulated incident and receive feedback on how their decisions can affect security, public safety, and service recovery.

2. Selected ODTN Scenario Description

In the selected scenario, a cyberattack disrupts a public transit agency's digital systems. Electronic schedules and ticketing services stop working correctly, and passenger information

becomes unreliable. Commuters experience delays and confusion, while transit employees face difficulties coordinating services.

The transit agency must respond quickly to reduce the impact of the incident. It needs to protect passengers, investigate the cause, preserve digital evidence, provide accurate public updates, and restore affected services safely.

This scenario is relevant to cybersecurity because public transportation is an essential service. A cyberattack can affect not only computer systems but also passenger safety, business continuity, public confidence, and the daily activities of an entire community.

3. Purpose and Learning Objectives

The purpose of TransitShield is to provide an interactive educational experience that demonstrates how cybersecurity professionals respond to a cyberattack affecting public transportation.

The main learning objectives are to:

- Understand the stages of cybersecurity incident response.

- Identify the importance of isolating affected systems and investigating suspicious network activity.

- Understand why digital evidence must be preserved during an investigation.

- Recognize the importance of protecting passengers and maintaining essential services.

- Learn how to communicate accurate information to management, employees, and the public.

- Understand why systems must be validated and restored from verified, clean backups.

- Identify security practices that can reduce the risk of future cyberattacks.

4. Website Features

4.1 Incident Dashboard

The incident dashboard presents an overview of the simulated cyberattack. It displays the incident status, affected transit services, security alerts, and the estimated number of commuters affected in the fictional scenario.

The dashboard helps users understand the situation before making decisions.

4.2 Incident Overview

This section introduces the ODTN scenario and explains how the cyberattack affects public transportation. It provides background information about the incident and allows users to start the simulation.

4.3 Interactive Crisis Response Simulator

The simulator allows users to act as members of a cybersecurity incident response team. It contains five decision stages in which users select actions to manage the incident.

The decisions include:

1. Isolating affected systems from the network to contain the threat.

2. Investigating suspicious network activity and preserving digital evidence.

3. Activating the incident response plan and protecting passengers.

4. Notifying management and providing accurate public updates.

5. Validating backups and restoring systems in stages.

After each decision, the simulator provides feedback explaining the possible consequences of the user's choice. This helps users understand which actions support effective incident response, and which actions may increase risks or delay recovery.

4.4 Attack Timeline

The attack timeline presents the fictional development of the incident, from the initial security alert to the disruption of transit operations, containment, investigation, and recovery.

It helps users understand how a cybersecurity incident develops and why each response stage is important.

4.5 Cybersecurity Response and Recovery

This section explains how an organization can respond to and recover from a cyberattack. The main steps include:

- Containment: Isolate affected systems to prevent the attack from spreading.**

- Investigation: Examine suspicious network activity and identify the possible cause of the incident.**

- Evidence preservation: Protect relevant logs and other digital evidence for analysis.**

- Eradication: Remove the threat and address the security weaknesses that allowed the attack.**

- Recovery: Validate backups, restore systems in stages, and test their functionality before reconnecting them.**

- Monitoring: Continue monitoring the network for suspicious activity after recovery.**

These steps demonstrate why incident response must be organized and carefully managed.

4.6 Security Assessment and Lessons Learned

This section presents important lessons from the simulated incident. It explains the need for regular security updates, access controls, network monitoring, tested backups, incident response planning, and employee cybersecurity awareness.

It also highlights the importance of reviewing an incident after recovery to identify weaknesses and prevent similar attacks in the future.

4.7 Cybersecurity Knowledge Quiz

The website includes a quiz to test users' understanding of cybersecurity incident response, evidence preservation, public communication, business continuity, and secure system recovery

Users can answer questions and review the correct solutions to improve their understanding of the subject.

4.8 Final Score and Feedback

At the end of the simulation, users receive a score based on their decisions. The feedback identifies actions that support effective incident response and explains which choices could increase security risks or delay service recovery.

This feature encourages users to reflect on their decisions and learn from the simulated outcomes.

5. Technologies and AI Tools Used

The project uses the following technologies and tools:

-HTML: Creates the structure and content of the website.

-CSS: Controls the website's appearance, layout, and visual design.

-JavaScript: Adds interactive features, decision-making options, feedback, scoring, and quiz functionality.

-ChatGPT: Provides assistance with project planning, content organization, and development ideas.

These technologies support the development of an interactive educational website that users can access through a web browser.

6. Security Considerations

TransitShield is designed as a client-side educational simulation. It does not require user accounts, collect personal information, or depend on external API keys.

The website uses simulated incident information and fictional outcomes for educational purposes. Its design aims to keep interactions simple and reduce unnecessary security risks.

Before submission, the website should be tested to confirm that navigation, decision buttons, scoring, quizzes, and feedback work correctly. The developer should also check for broken links, incorrect information, and other functionality issues.

The project emphasizes that real organizations should follow an established incident response plan, preserve digital evidence, protect sensitive information, and verify that systems are safe before restoring normal operations.

7. Cybersecurity Concepts Demonstrated

The TransitShield project demonstrates several important cybersecurity concepts:

Incident response: Taking organized action to detect, contain, investigate, and recover from a cybersecurity incident.

Network security: Identifying suspicious network activity and isolating affected systems to reduce the spread of an attack.

Risk management: Evaluating possible consequences before choosing a response action.

Evidence preservation: Protecting logs and other digital evidence to support an investigation.

Business continuity: Maintaining essential transit operations and protecting passengers while systems are disrupted.

Disaster recovery: Restoring affected systems from verified, clean backups and confirming that they work safely.

****Security awareness: Teaching users how appropriate decisions and preventive security practices can reduce the impact of cyberattacks.**

The project also highlights the importance of cooperation between cybersecurity specialists, IT staff, transit operations, management, and public communication teams.

8. Practical Demonstration and Screenshots

The practical demonstration shows the TransitShield website and its interactive features.

See relevant screenshots below:

1. Screenshot 1: Main website or incident dashboard.

2. Screenshot 2: Selected ODTN scenario and incident overview.

3. Screenshot 3: Interactive crisis simulator and decision-making options.

4. Screenshot 4: Attack timeline or incident response and recovery section.

5. Screenshot 5: Cybersecurity knowledge quiz and questions.

6. Screenshot 6: Quiz solutions, final score, and feedback.

These screenshots provide evidence of the website's design and the interactive features developed for the assignment.

Website file: `index.html`

C:\Users\Ralph\OneDrive\Documents\index.html

file:///C:/Users/Ralph/OneDrive/Documents/index.html

***Note: This is a local computer file path. from my PC, it is not a public website link.**

9. Expected Outcome

The expected outcome is an interactive educational website that simulates a cyberattack affecting public transportation. The project helps students understand the real-world effects of cyberattacks on essential services and learn how cybersecurity professionals make decisions during an incident.

Through the simulator, users can practise evaluating security alerts, containing threats, protecting passengers, communicating with the public, and restoring services safely.

The project also demonstrates how HTML, CSS, JavaScript, and AI-assisted planning can be combined to create an educational cybersecurity experience.

10. Conclusion

TransitShield demonstrates that responding to a cyberattack involves more than fixing computer systems. Organizations must protect people, maintain essential services, investigate the incident, communicate honestly, and restore systems carefully.

By using an interactive simulation based on the ODTN scenario "Cyberattack Cripples Transit – Commuters Stranded," this project connects cybersecurity theory with practical decision-making. It helps users understand how effective incident response, network protection, business continuity, and secure recovery can reduce disruption and improve organizational resilience.

Overall, the project provides an educational way to explore cybersecurity challenges affecting public transportation and understand the importance of making safe, informed decisions during a crisis.

Thanks